

Birdcam Python Coding Standards

Language: *Python*

Comment Rules

1. In every file, there must be a comment at the top of the file stating its name, purpose, author, and date.

Example:

```
"""
convert.py
Converts the pseudocode to a flowchart
```

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Date: 22 Sep 2025

```
"""
```

2. Every function/class must have a comment describing its purpose, arguments of the function, and the return values.

Example:

```
"""
Read the pseudocode file to display
Arg:
    file_name - name of the input file
"""
def read_file(file_name):
```

3. Every variable must have a brief description of its purpose

Example:

```
bool correct_format = False # check if the format of the pseudocode is correct
```

4. Explain the code where it is hard to understand with comments

Naming rule - _case

1. The name of the variable and function must be meaningful. For example:

```
def get_shape:
    int index
```

2. The variable name should all be lowercase
3. For variable names that have multiple words use snake case.

For example:

```
string selected_shape
```

4. Avoid too short or long variable name

5. Always state the data type of the variable.

For example:

```
int index
```

Indentation

1. Use 4 spaces per indentation level.
2. Avoid extraneous whitespace in the following situations:
 - Immediately inside parentheses, brackets or braces:

Correct:

```
spam(ham[1], {eggs: 2})
```

Wrong:

```
spam( ham[ 1 ], { eggs: 2 } )
```

- Between a trailing comma and a following close parenthesis:

Correct:

```
foo = (0,)
```

Wrong:

```
bar = (0, )
```

- Immediately before a comma, semicolon, or colon:

Correct:

```
if x == 4: print(x, y); x, y = y, x
```

Wrong:

```
if x == 4 : print(x , y) ; x , y = y , x
```